

Wall Creation

in Revit

Basic | Stacked | Curtain | Layers | Joins | Properties

What You Will Learn

- Three wall categories: Basic, Stacked, and Curtain Wall — when to use each
- Wall families and types — how Revit organises and stores wall data
- Drawing walls: the Wall tool, options bar, and constraint settings
- Compound wall layers: structure, functions, materials, and core boundaries

Wall Properties Example

Base: Level 1
Top: Level 2
Height: 3,000 mm
Type: Brick 215mm

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REVIT-005: Wall Creation in Revit

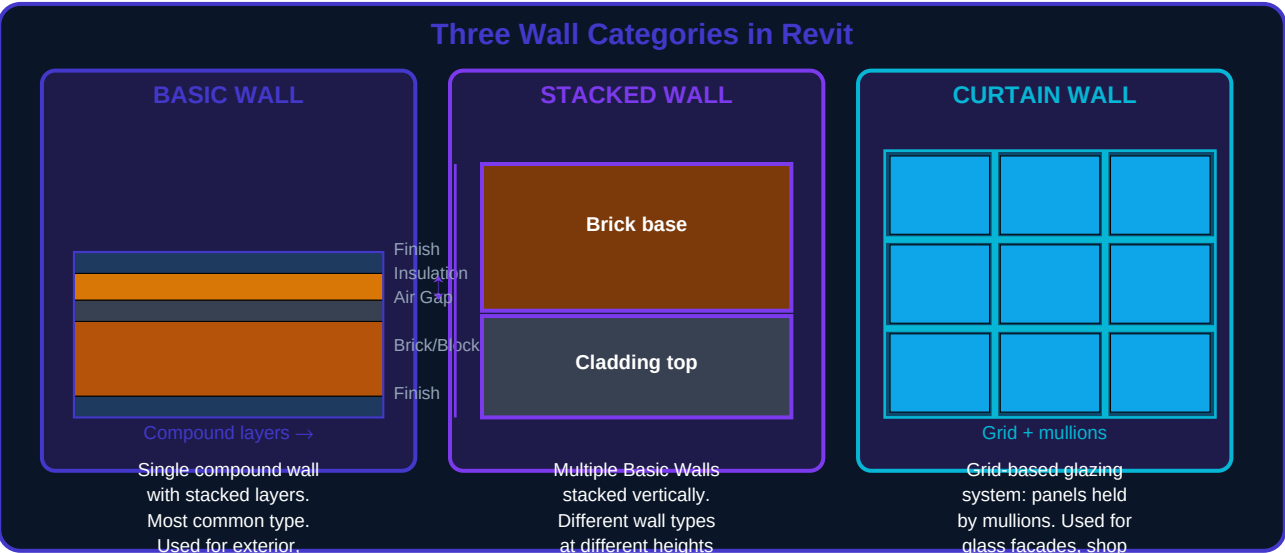
Walls are the primary building element in every Revit architectural model. They define space, support structure, carry MEP services, and form the visual character of a building. Unlike walls drawn as simple lines in AutoCAD, Revit walls are intelligent objects: they know their height, material composition, relationship to adjacent elements, and structural role. Understanding how to create, configure, and edit walls correctly is one of the most important foundational skills in Revit.

Learning Objectives

- Identify and use the three wall categories: Basic Wall, Stacked Wall, and Curtain Wall
- Navigate the wall type library, duplicate existing types, and create new wall types
- Draw walls using the Wall tool with correct constraint, height, and location line settings
- Edit compound wall structure: add, remove, and reorder layers and assign materials
- Control wall joins using Revit's automatic and manual join cleanup tools

1. The Three Wall Categories

Revit organises all walls into three fundamental categories. Choosing the correct category before drawing is important — you cannot convert a Basic Wall to a Curtain Wall after placing it.



Category	When to Use	Key Characteristic
Basic Wall	Standard exterior and interior walls of any material — brick, concrete, timber frame, glass block	Composed of parallel compound layers with defined functions and materials. The most versatile type.

Stacked Wall	When a wall changes material or thickness at a specific height — e.g. brick veneer below sill, metal cladding above	A "wrapper" type that stacks two or more Basic Wall types vertically at defined offsets.
Curtain Wall	Glass facade systems, shop fronts, and any gridded panel/mullion system regardless of panel material	Grid-based system with separate panel and mullion families. Panels can be glass, solid, or any material.

2. Wall Families and Types

In Revit, every wall belongs to a **Family** (Basic Wall, Stacked Wall, or Curtain Wall) and a **Type** within that family. The Type defines the specific construction — "Exterior - Brick 215mm with Plasterboard" or "Interior - Partition 100mm Stud." Types are stored in the project and can be duplicated and modified without affecting other instances.

Term	Description	Example
Family	The broad category of the wall element. Set when inserting.	Basic Wall (covers all non-curtain-wall types)
Type	The specific wall construction within the family. Each type has its own layer structure, materials, and dimensions.	"Exterior - Brick on Mtl. Stud", "Interior - 4" Partition", "Foundation - 12" Concrete"
Instance	A single placed wall in the model. Each instance can have unique Base and Top constraints, height offsets, and structural usage.	The south wall on Level 1 — same type as all other south walls but its own position and height.
Type Properties	Settings shared by ALL instances of a type: layer structure, total thickness, function, thermal mass.	Changing the insulation thickness in Type Properties updates every wall of that type in the whole model.
Instance Properties	Settings unique to ONE placed wall: Base Level, Top Level, Base Offset, Top Offset, Location Line.	Only the south wall has an extra 200mm base offset because it sits on a raised plinth.

3. Drawing Walls — The Wall Tool

The Wall tool is accessed via Architecture tab → Build panel → Wall (keyboard shortcut: **WA**). Before drawing, always configure the Options Bar and Properties panel — drawing with incorrect settings is the most common cause of walls that require extensive correction later.

Setting	Location	What to Set Before Drawing
Wall Type	Type Selector (Properties panel, top)	Select the correct wall type. If the right type doesn't exist, duplicate an existing similar type and edit it.
Height	Options Bar → Height / Unconnected	Set "Top Constraint" to the upper level (e.g. Level 2) for storey-height walls, or set "Unconnected Height" in mm for walls not reaching a level.

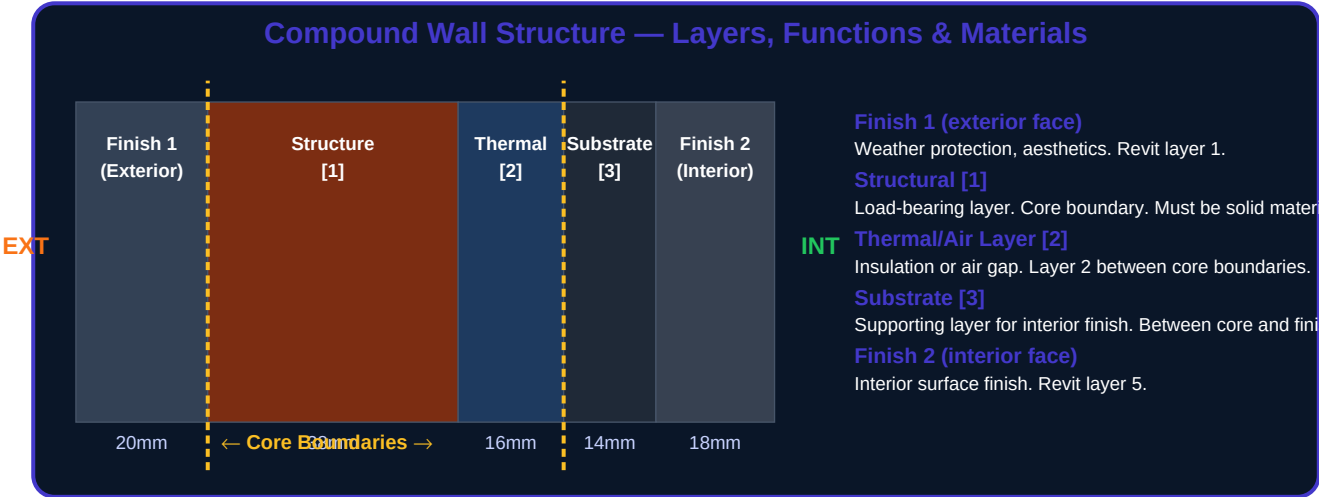
Location Line	Options Bar → Location Line	Sets which line of the wall you are drawing on: Wall Centreline, Core Centreline, Finish Face (Interior/Exterior), or Core Face. For exterior walls: use Finish Face Exterior so the outer surface aligns to your reference lines.
Chain	Options Bar → Chain checkbox	Tick to draw multiple connected wall segments in one operation (endpoint of each segment becomes the startpoint of the next).
Offset	Options Bar → Offset	Draws the wall offset from the cursor line by a specified distance. Useful for offsetting from a gridline to the wall face.

4. Wall Properties — Height and Constraints



5. Compound Wall Structure — Layers, Functions, Materials

The defining feature of a Revit Basic Wall is its compound layer structure. Each layer has a function, material, and thickness. Layers are organised relative to two **Core Boundaries** — the structural heart of the wall. Layers outside the core are finishes and substrates; layers inside are structural and thermal elements.



Layer Functions in Revit

Function	Prior ity	Role	Revit Cleanup Rule
Structure [1]	1 (high est)	The load-bearing core of the wall. Structural elements like concrete or brick. Defines the "core" boundaries.	Structure layers connect to matching Structure layers at wall joins (T-joins, corners) to maintain structural continuity.
Substrate [2]	2	Support for other layers. Usually a board or rigid backing (e.g. OSB, plasterboard over studs).	Substrate layers wrap around Structure layers at joins.
Thermal / Air Layer [3]	3	Insulation or ventilated air cavity. Zero-thickness air layers can represent cavities.	These layers are trimmed at joins but do not connect through.
Membrane Layer [4]	4	Vapour barrier, DPC, or waterproofing membrane. Typically zero thickness.	Treated as a Substrate in join calculations.

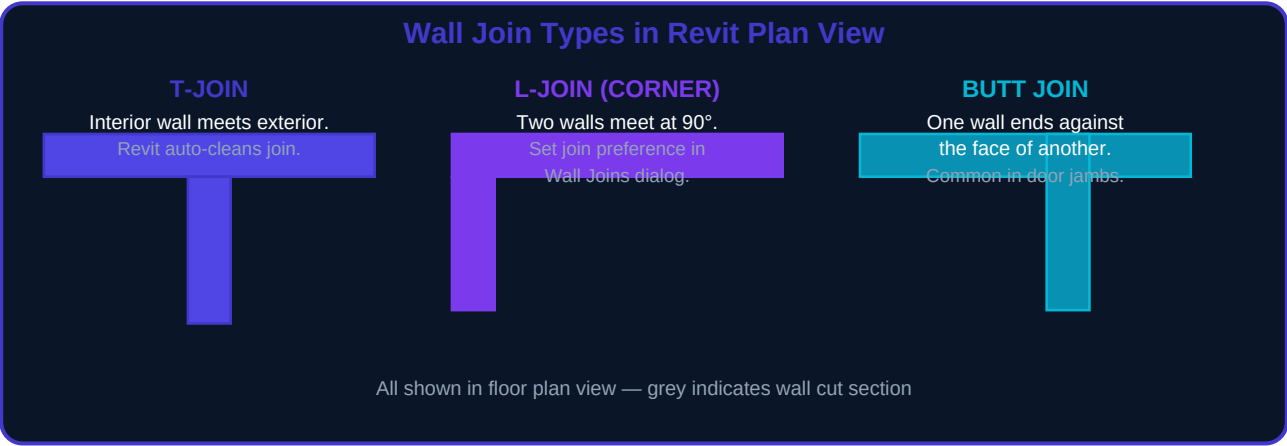
Finish 1 [5] / Finish 2 [5]	5 (lo west)	Exterior (Finish 1) and interior (Finish 2) surface materials — plaster, render, tiles, paint.	Finish layers are trimmed at wall ends and joins. They do not pass through other walls.
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Editing Wall Type Layers — Step by Step

S t e p	Action	Where in Revit
1	Select any wall of the type you want to edit	Click wall in model view or floor plan
2	Open Type Properties	Properties panel → Edit Type button (or keyboard: ET)
3	Click "Duplicate" to create a new type	Always duplicate before editing — never modify shared types directly
4	Name the new type clearly	Use format: "Exterior - Material - Thickness" e.g. "Exterior - Brick 215mm Cavity Insul"
5	Click "Edit..." next to Structure parameter	Opens the Edit Assembly dialog with the layer table
6	Add/remove/reorder layers, assign materials, set thicknesses	Use Insert, Delete, Up, Down buttons. Material column links to Material Browser.
7	Verify total thickness shown at bottom	Cross-check against specification or architectural drawings
8	Click OK twice to apply	All walls of this new type update immediately throughout the project

6. Wall Joins — Automatic and Manual Cleanup

When two Revit walls meet, Revit automatically calculates how the layers should join and clean up the intersection. The automatic join is governed by layer functions: Structure layers join to Structure, Finish layers trim at the junction. In most cases automatic join works well, but complex multi-material corners may need manual cleanup.



Join Tool / Setting	How to Access	What It Does
Allow Join (default)	Modify tab → Geometry panel → Wall Joins	Walls that touch automatically join and clean up layers. The standard mode for most walls.
Disallow Join	Right-click wall end → Disallow Join	Prevents a wall end from joining to adjacent walls. Use when you want a wall to butt up without cleanup (e.g. temporary partitions, or when automatic join is wrong).
Wall Joins dialog	Modify tab → Geometry → Wall Joins → click a join	Opens the Wall Joins dialog for a specific intersection. Allows you to cycle through join options: Butt (which wall wins), Miter (45° mitre cut), or None.
Switch Join Order	Right-click join → Switch Join Order	In a T-join or L-join, determines which wall's structure passes through and which wraps around. Essential for correct structural representation.
Trim/Extend to Corner	Modify tab → TR or TE shortcut	Trims or extends two walls to meet precisely at a corner. Essential for cleaning up walls that were drawn too long or short.

7. Structural vs Architectural Walls

Every Revit wall has a **Structural Usage** property that determines how the wall behaves in structural analysis, reinforcement tagging, and load calculations.

Structural Usage	When to Use	Impact on Project
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Non-Bearing (default)	Interior partitions, cladding, non-structural finishes	Treated as architectural element only. Not included in structural analysis models.
Bearing	Load-bearing masonry walls, reinforced concrete walls, structural block walls	Included in structural analysis. Can have rebar placed in them. Exported to structural analysis software.
Shear	Concrete or masonry walls designed to resist lateral (wind/seismic) loads	Treated as lateral force resisting element in structural model.
Structural Combined	Walls acting as both bearing and shear walls simultaneously	Combined structural role — most common for reinforced concrete walls in multi-storey buildings.
Retaining	Basement and foundation walls resisting soil pressure	Special structural classification for walls in contact with earth. Separate design requirements.

8. Practice Exercises

#	Exercise	Step-by-Step Instructions
1	Draw a simple room in plan	1. Open a new Revit project from an architectural template. 2. Use Wall tool (WA) to draw a 5×4m rectangular room with 200mm interior partition walls. 3. Set Base = Level 1, Top = Level 2 (or 3000mm unconnected). Verify in 3D view.
2	Create a custom exterior wall type	1. Select any existing exterior wall → Edit Type → Duplicate. 2. Name it "Exterior - Brick 215mm + 100mm Insulation + Plasterboard". 3. In Edit Assembly: add layers with correct functions, thicknesses, and Revit materials. Verify total = 340mm.
3	Curtain wall on a facade	1. Draw an exterior wall → change its type to Curtain Wall from the Type Selector. 2. In Type Properties: set vertical grid spacing = 1000mm, horizontal = 800mm. 3. Add mullion type: Rectangular 50×150mm. View result in perspective view.
4	Fix wall joins	1. Draw an L-shaped room with two exterior walls meeting at a corner. 2. Zoom into the corner in floor plan view and observe automatic join cleanup. 3. Right-click the join → Switch Join Order. Observe which wall structure now dominates.
5	Stacked wall exercise	1. Create a stacked wall type with 600mm brick base (up to 1000mm height) and lightweight partition above. 2. Place this wall in the model and verify it shows correctly in section view. 3. Tag the wall with a Wall Tag annotation and verify it shows the correct type name.

Lesson Summary

Key Concept	Summary Rule
Wall categories	Basic = standard compound layers. Stacked = two Basic walls at different heights. Curtain = grid/mullion/panel glazing system.
Type vs Instance	Type Properties (layers, materials, thickness) → affects ALL walls of that type. Instance Properties (Base, Top, Height, Location Line) → affects ONE placed wall only.
Drawing walls	Always set: Type, Height/Constraint, and Location Line BEFORE drawing. WA = keyboard shortcut. Chain = draw connected segments.
Location Line	For exterior walls: Finish Face Exterior. For interior partitions: Wall Centreline. This controls which edge of the wall aligns to your sketch reference.
Compound layers	Structure [1] = core/bearing. Substrate [2] = backing board. Thermal [3] = insulation. Finish [5] = surface material. Core Boundaries separate structural from finish layers.
Wall joins	Automatic join works in most cases. Use Wall Joins dialog for complex corners. Switch Join Order to control which wall's structure dominates at intersections.

Structural Usage	Set to "Bearing" for load-bearing masonry/concrete. "Non-Bearing" for partitions. Affects structural analysis model export and rebar placement.
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What Is Next — REVIT-006

REVIT-006: Floor Creation — Creating and editing floor slabs in Revit. Floor types and compound structure. Picking walls to define floor boundaries. Slope arrows and sloped floors. Floor openings and floor-to-wall joins. Structural floor vs architectural floor. Floor properties and scheduling.

Resource	Details
Revit PDF Series	REVIT-001 through REVIT-030 — ayetechub.com/pdfs.html
Revit Downloads	Free student licence — autodesk.com/education
Community	Telegram: t.me/ayetechub